

Drone that can catch another by firing a net at it



Drone catches another by firing a net at it (Image courtesy: www.brunchnews.com)

Engineers from Human-Interactive Robotics Lab (HIROLab) at Michigan Technological University have filed a patent for a prototype for a drone-catching system that fires a net to take other unwanted aircraft down.

They have named the project Robotic Falconry and have said that the drone, equipped with a net shooter, can intercept and physically remove any intruding multi-rotor drone from private airspace. It can be autonomous or remote-controlled while tackling a drone.

According to the researchers, the net-shooting technique can be effective when force-landing unmanned intruders that would otherwise put the public at risk.

Drone that can do donuts, figure-eights around obstacles

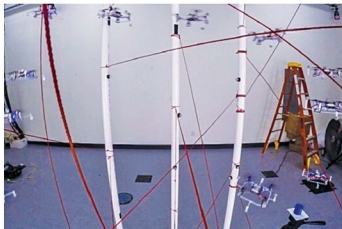
Getting drones to fly around without hitting things is a huge task. Obstacle-detection and motion-planning are two of computer science's trickiest challenges, because of the complexity involved in creating real-time flight plans that avoid obstacles and handle surprises like wind and weather.

Two teams of researchers from MIT's Computer Science and Artificial Intelligence Laboratory (CSAIL) have developed software that allows drones to stop on a dime to make hairpin movements over, under and around some 26 distinct obstacles in a simulated forest.

One team has shown a small quad-rotor doing donuts and figure-eights through an obstacle course of strings and PVC pipes.

In a second CSAIL project, PhD student Anirudha Majumdar showed off a fixed-wing plane that avoids obstacles without any advance knowledge of the space, and even in the face of wind gusts and other dynamics. The approach was to pre-program a library of dozens of

distinct funnels that represent the worst-case behaviour of the system, calculated via a rigorous verification algorithm.



Researchers are teaching drones to fly autonomously without ever crashing (Image courtesy: www.news.mit.edu)